

Personal Knowledge OS (PKOS)

Product Requirements Document (PRD)

1 Vision

Personal Knowledge OS is an AI-powered learning and knowledge management platform that helps users capture information from multiple sources, organize it into a structured knowledge base, retrieve it intelligently, and retain it through personalized revision recommendations.

The goal is not simply to store information, but to transform scattered content into long-term, reusable knowledge.

2 Problem Statement

Modern learners consume information from many sources:

- Articles
- Documentation
- YouTube videos
- Research papers
- Personal notes
- Tutorials

Over time, several problems emerge:

1. Information becomes scattered across bookmarks, notes, PDFs, and browser tabs.
2. Users remember learning something but cannot locate it quickly.
3. Related concepts remain disconnected.
4. Valuable knowledge is forgotten due to lack of revision.
5. Existing note-taking tools focus on storage rather than learning and retention.

As a result, users repeatedly relearn concepts they have already studied.

3 Target Users

3.1 Primary Users

3.1.1 Students

- Engineering students
- Competitive programmers
- University learners

Need:

- Organize study materials
- Revise concepts efficiently
- Track learning progress

3.1.2 Software Engineers

Need:

- Store technical knowledge
- Quickly retrieve information during development
- Build a personal technical knowledge base

3.1.3 AI/ML Learners

Need:

- Manage research papers
- Connect concepts across topics
- Generate summaries and revision material

4 Product Goals

The platform should help users:

1. Capture knowledge from any source.
2. Organize information automatically.
3. Retrieve knowledge instantly.
4. Discover relationships between concepts.
5. Revise information before forgetting it.
6. Build a long-term personal knowledge repository.

5 Core Product Pillars

5.1 Pillar 1: Knowledge Capture

The system should allow users to save information from multiple sources.

Supported sources:

- Web articles
- PDFs
- YouTube videos
- Manual notes
- Documentation pages

For every saved source, the platform stores:

- Title
- Content
- Source URL

- Date added
- Source type
- User metadata

5.2 Pillar 2: Knowledge Organization

The system automatically structures saved content.

Each item should be categorized using:

Domain

- Programming
- AI
- Databases
- System Design
- Networking

Topic

- Redis
- PostgreSQL
- Docker
- RAG

Difficulty

- Beginner
- Intermediate
- Advanced

Content Type

- Article
- Video
- PDF
- Note

Status

- Unread
- Reading
- Learned
- Mastered

The user may manually edit classifications.

5.3 Pillar 3: Knowledge Retrieval

Users should be able to search their knowledge base naturally.

Example queries:

- What do I know about Redis?
- Show all notes about authentication.

- Find resources related to system design.
- Which articles discuss caching?

The platform should return:

- Relevant sources
- Extracted snippets
- Related concepts
- AI-generated summaries

5.4 Pillar 4: AI Question Answering

Users can ask questions against their knowledge base.

Examples:

- Explain Redis based on my saved resources.
- Summarize everything I know about databases.
- Compare PostgreSQL and MongoDB using my notes.

Responses must:

- Use only saved content.
- Cite sources.
- Link back to original documents.

If information is unavailable, the system should clearly indicate:

“This information does not exist in your knowledge base.”

5.5 Pillar 5: Knowledge Connections

The system should automatically identify relationships between content.

Example:

Article A:

- Redis caching

Article B:

- System design interview notes

Detected connection:

- Scalability
- Performance optimization

Users should be able to explore:

- Related documents
- Related topics
- Related concepts

5.6 Pillar 6: Revision & Retention

The system should actively help users retain knowledge.

Each knowledge item maintains:

- Date added
- Last revision date
- Revision count
- Confidence score
- Knowledge score

5.6.1 Revision Recommendation Engine

The system recommends revisions based on:

- Time since learning
- Time since last revision
- Confidence level
- Topic importance
- User activity

Example:

Today's Revision List:

1. Redis Fundamentals
2. Bellman-Ford Algorithm
3. HTTP Caching

5.6.2 Smart Revision Material

Instead of reopening entire documents, the system generates:

- Key points
- Summaries
- Flashcards
- Quiz questions
- Important concepts

All generated from saved knowledge.

5.7 Pillar 7: Learning Progress Tracking

The platform tracks learning activity.

Metrics:

- Total knowledge items
- Topics learned
- Revision streak
- Most studied domains
- Weakest knowledge areas
- Knowledge growth over time

6 User Dashboard

The home dashboard should display:

6.1 Knowledge Overview

- Total saved resources
- Total topics
- Total revisions completed

6.2 Today's Revision Queue

Recommended items for revision.

6.3 Recently Added Knowledge

Recent articles, notes, and PDFs.

6.4 Learning Insights

Examples:

- Networking has not been revised in 30 days.
- Databases is your most studied topic.
- You are currently learning Backend Development.

7 Search Experience

The platform should support:

7.1 Structured Search

Examples:

- Topic = Databases
- Status = Learned
- Difficulty = Intermediate

7.2 Semantic Search

Examples:

- How can I improve backend performance?
- Explain caching strategies.

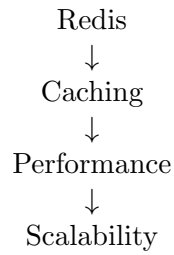
8 Knowledge Graph View (Future)

Visual representation of relationships between:

- Topics
- Concepts

- Documents

Example:



Users can explore how concepts connect across their knowledge base.

9 Success Metrics

The product is successful if users can:

1. Find previously learned information within seconds.
2. Reduce repeated relearning.
3. Build a structured long-term knowledge repository.
4. Maintain a consistent revision habit.
5. Discover new connections between saved knowledge.

10 Product Vision Statement

Personal Knowledge OS transforms information into retained knowledge by combining intelligent organization, semantic retrieval, AI-assisted understanding, and personalized revision guidance within a single learning platform.